

Nishant Sharma

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Education

New York University, Tandon School of Engineering · New York, NY Master of Science in Computer Engineering · GPA: 3.7	May 2026
Dr. A.P.J. Abdul Kalam Technical University · Delhi, India B.Tech Computer Science and Engineering · First Division with Distinction	Jun 2024

Technical Skills

Languages	Python · Java · TypeScript · JavaScript · SQL
Backend	FastAPI · Node.js · REST APIs · WebSockets · Redis · Celery · ARQ · SSE
Databases	PostgreSQL · MySQL · MongoDB · AWS RDS
Cloud/Infra	Linux · AWS · Docker · Kubernetes · Terraform · Jenkins · GitHub Actions
AI/ML	LLM APIs · RAG pipelines · ONNX Runtime · vector search · PyTorch
Frontend	React · Next.js · Tailwind

Work Experience

Lead Software Engineer · Ingelt Study Abroad	Sep 2023 – Aug 2024
- Owned backend architecture for a 9-engineer team, including API design, PostgreSQL schema decisions, sprint planning, code reviews, and production release workflows. - Designed a unified Node.js API and PostgreSQL schema serving students, teachers, partners, and internal staff, replacing three redundant backends. - Architected AWS deployment using EC2, ALB, RDS, SQS, S3, and CloudFront with async queues and DLQs for email and video processing. - Built Jenkins CI/CD from scratch, automating test, build, and EC2 deployment workflows.	
Co-Founder · Macverin Technologies · macverin.com	Jul 2022 – Aug 2023
- Delivered 12 client platforms in 14 months across ride-hailing, booking, visa consultancy, and internal operations domains. - Built role-based application processing software for an 8-person visa operations team using Node.js, FastAPI, PostgreSQL, and Firebase Auth. - Integrated Stripe, Google Maps APIs, Firebase Auth, and AWS deployments across client platforms with isolated business logic per customer. - Designed a client-agnostic fare calculation layer separating pricing rules from Google Maps API responses, enabling pricing updates without integration changes.	

Projects

Singularity - AI Deep Research Platform · singularity.hellonish.dev	Mar 2026 – Apr 2026
- Implemented hybrid retrieval with vector search and reranking to ground agent outputs in source documents, improving citation quality and reducing irrelevant context passed to downstream agents. - Built a multi-agent research platform using FastAPI, Redis, and DAG-based task orchestration, enabling researcher agents to execute subtasks across 44 modular skills. - Implemented Redis sorted-set rate limiting with atomic pipelines to enforce per-user request and token budgets while eliminating fixed-window burst artifacts. - Reduced embedding worker memory footprint by 94% by replacing PyTorch sentence-transformers with ONNX Runtime fastembed on the same MiniLM model. - Designed auto-registration for skills using filesystem discovery and class-level registration, reducing new skill integration to a single file change.	
Wand - AI Career Intelligence Platform · wand.hellonish.dev	Jan 2026 – Apr 2026
- Built an AI job intelligence pipeline that reduced manual company/JD research from 30–60 minutes to a single automated run. - Designed an asyncio DAG scheduler to execute independent research steps in parallel, reducing pipeline wall-clock latency by 50%. - Streamed real-time pipeline progress from Celery workers to the browser using Redis pub/sub and WebSockets, avoiding client polling. - Implemented deterministic profile merging using hash-based deduplication and stable sorting, with LLM normalization to resolve entity mismatches.	
PGDR - Persistent Gradient Disagreement Reweighting · paper	Jan 2026 – May 2026
- Proposed PGDR, a pseudo-group discovery method that uses last-layer gradient geometry to identify shortcut-conflicting examples without group labels. - Improved Waterbirds worst-group accuracy from 68.38% to 81.52% while preserving 92.56% average accuracy, validated across 3 seeds with permutation p=0.0002. - Ran ablations showing warmup quality as the key failure mode: weak warmup reduced WGA to 66.51%, while naive pseudo-group refresh degraded selection quality to AUC 0.465.	

Teaching Experience

Machine Learning Teaching Assistant · NYU Tandon School of Engineering, New York, NY	Sep 2025 – May 2026
- Supported office hours and assignment feedback for graduate ML coursework, helping students debug model training, tensor shape mismatches, and reproducibility issues. - Led debugging sessions for graduate students on PyTorch training pipelines, covering optimization, autograd, data loading, and environment failures. - Standardized PyTorch assignment templates and troubleshooting guides, reducing environment-related debugging overhead during office hours.	